

iRODS as a composable building block for data management at SURF

Francisco Morales - SURF DMS team

SURF data management current situation

- Serves Dutch universities and research institutions with innovative (IT) solutions to increase research impact
- Different research communities with different (data) needs
- Long-term (data) projects as well as short-lived PoCs
- DMS team has experience with iRODS and Yoda, serving 7+ research institutions

Challenge

- Move from dedicated, rigid computing environments for fixed needs toward robust workflow components that can be reused and composed to serve different needs in a timely manner

Proposed solution

- Provide iRODS as a composable building block via containerization on K8s: CI/CD pipeline integration; HA; auto-scaling, among other goodies

Obstacles

- Hostname attached to the irods container = hinder HA
- Data transfer dies = intermediate replicas that need to be handled manually